

Inter-Process Communication

Inter-process communication, or IPC, is the way separate processes on a system pass data to each other and keep their actions in sync. Because the OS keeps every process in its own protected memory, they can't just read each other's variables, so it provides channels like pipes, message queues, shared memory, semaphores, signals, and sockets to let them talk.

There are 8 IPC techniques in which 2 are as follows:

1. Message queues

Message Queue is a list of messages stored in the operating system kernel. One process sends messages to the queue and another process reads them in order, so both do not need to run at the same time.

2. Sockets

Socket is an endpoint for two-way communication between processes. Sockets allow processes to exchange data on the same computer or across different computers over a network.

Example: Imagine an online food ordering system where the customer's app sends an order to the restaurant server using a socket. The server places the order in a message queue, and the kitchen process reads and prepares the orders one by one.

For a message queue, the server and kitchen work independently. For a socket, the app and server communicate over the network in real time.